

MINOR PROJECT

STATIC WEBSITE HOSTING USING EC2 AND S3 (NGINX)

Implementation of a Static Website using Amazon EC2, Nginx and Amazon S3

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Project: Static Website Hosting Using EC2 and S3

Platform: Amazon Web Services

Server: Nginx

Operating System: Amazon Linux 2023

Submission Date: 04 September 2026

1. Introduction

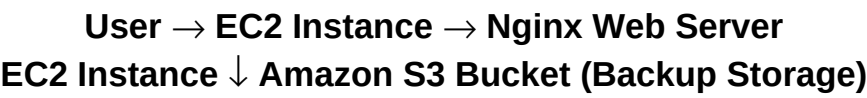
Amazon EC2 provides virtual servers that can be used to host web applications and websites. In this project, an EC2 instance running Amazon Linux 2023 was configured with the Nginx web server to host a static website.

Amazon S3 was used as backup storage for the website files. The website was accessed through the public IPv4 address of the EC2 instance.

2. Objectives

- Launch and configure an Amazon EC2 instance.
- Install and configure the Nginx web server.
- Create and host a static HTML website.
- Configure Security Group rules for web access.
- Create an Amazon S3 bucket.
- Upload the website backup file to S3.
- Verify the website through the EC2 public IP.
- Understand the basic use of EC2, Nginx and S3.

3. Project Architecture



Component	Purpose
Amazon EC2	Hosts the website
Amazon Linux 2023	Operating system
Nginx	Web server
Security Group	Controls network access
Amazon S3	Stores website backup
HTML	Website content

4. EC2 Instance Configuration

An EC2 instance was launched using Amazon Linux 2023.

Setting	Configured Value
Instance Type	t3.micro
Instance State	Running
Availability Zone	ap-south-1b
Public IPv4	13.127.119.52
Private IPv4	172.31.11.217
Web Server	Nginx

Implementation Evidence: EC2 instance is running and configured for the website.

5. Security Group Configuration

The Security Group was configured to allow incoming traffic required for the website and remote administration.

Type	Protocol	Port	Source
HTTP	TCP	80	0.0.0.0/0
SSH	TCP	22	0.0.0.0/0

Port 80 allows HTTP access to the website. Port 22 allows SSH access to the EC2 instance.

Note: For a real deployment, SSH access is normally restricted to a trusted IP rather than 0.0.0.0/0.

6. Nginx Web Server Installation

Nginx was installed on the Amazon Linux 2023 EC2 instance using the following commands:

```
sudo dnf update -y
sudo dnf install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
```

The Nginx service was checked using:

```
sudo systemctl status nginx
```

The service status showed **Active: active (running)**, confirming that the Nginx web server is running successfully.

7. Static Website Creation

The website was created as an HTML file named **index.html** and placed in the Nginx web root:

```
/usr/share/nginx/html/index.html
```

The website contains the project heading, AWS EC2 + Nginx + S3 information, project description, server information and operating system information.

The website was verified through the EC2 public IPv4 address. The displayed page confirms that the static website is being served by Nginx.

8. Amazon S3 Backup

An Amazon S3 bucket was created to store a backup copy of the website.

Bucket name: dalia-static-website-backup-2026-170638969506-ap-south-1-an

The website file **index.html** was uploaded to the bucket.

Object	Type	Storage Class
index.html	HTML	Standard

9. IAM Role Configuration

An IAM role named **StaticWebsiteS3Role** was associated with the EC2 instance. The role provides permissions required for the EC2 instance to interact with AWS services.

Implementation Evidence: IAM role and its attached policies are shown in the collected AWS console screenshot.

10. Verification Results

Component	Verification	Result
EC2	Instance running	Successful
Security Group	HTTP/SSH rules configured	Successful
Nginx	systemctl status nginx	Active / Running
Website	Accessed through public IP	Successful
S3	index.html uploaded	Successful
IAM Role	Role attached to EC2	Configured

11. Project Deliverables

- EC2 Instance Configuration screenshot
- Security Group Configuration screenshot
- Nginx Service Status screenshot
- Hosted Static Website screenshot
- S3 Bucket screenshot
- Uploaded index.html screenshot
- IAM Role Configuration screenshot

12. Learning Outcomes

- Basic AWS EC2 instance management
- Amazon Linux 2023
- Nginx web server installation
- Static website hosting
- Security Group configuration
- HTTP and SSH ports
- Amazon S3 storage
- Website backup using S3
- IAM roles and permissions
- Accessing an EC2-hosted website using a public IP

13. Conclusion

The project demonstrates static website hosting using Amazon EC2 and Nginx, with Amazon S3 used for storing a backup copy of the website.

An EC2 instance running Amazon Linux 2023 was configured with Nginx, the HTML website was created and accessed through the EC2 public IP, and the index.html file was uploaded to an S3 bucket for backup storage.

The collected screenshots provide evidence of the EC2 configuration, Security Group, Nginx service, working website, S3 bucket and uploaded website file.

14. Implementation Evidence — Screenshots

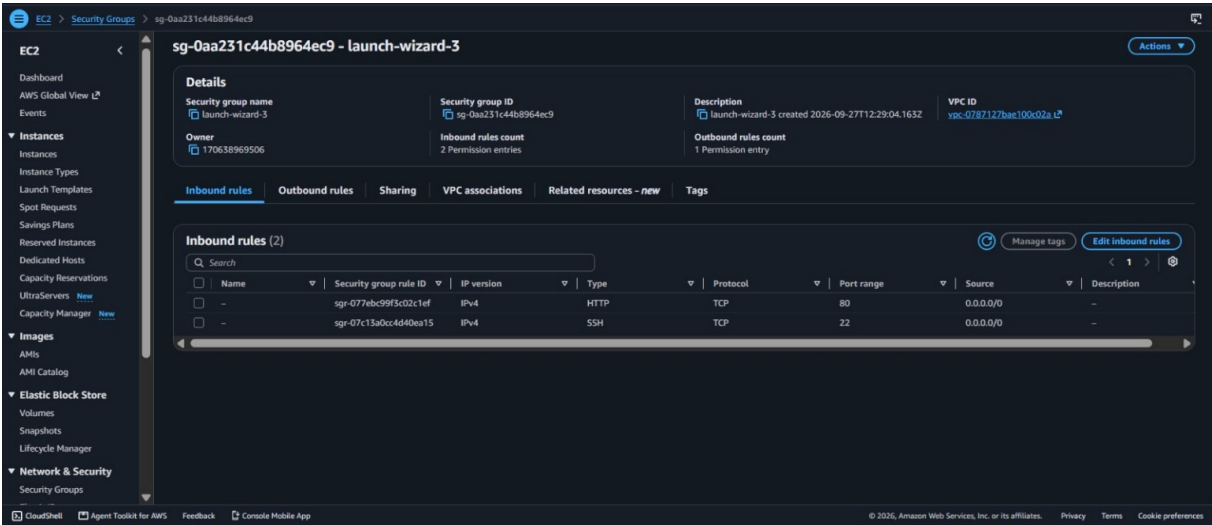


Figure 1: EC2 Security Group configuration

Welcome to My Static Website

AWS EC2 + Nginx + S3

This website is hosted on an Amazon EC2 instance using Nginx.

Website backup files are stored in Amazon S3.

Project: Static Website Hosting Using EC2 and S3

Server: Nginx

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Figure 2: Nginx-hosted static website

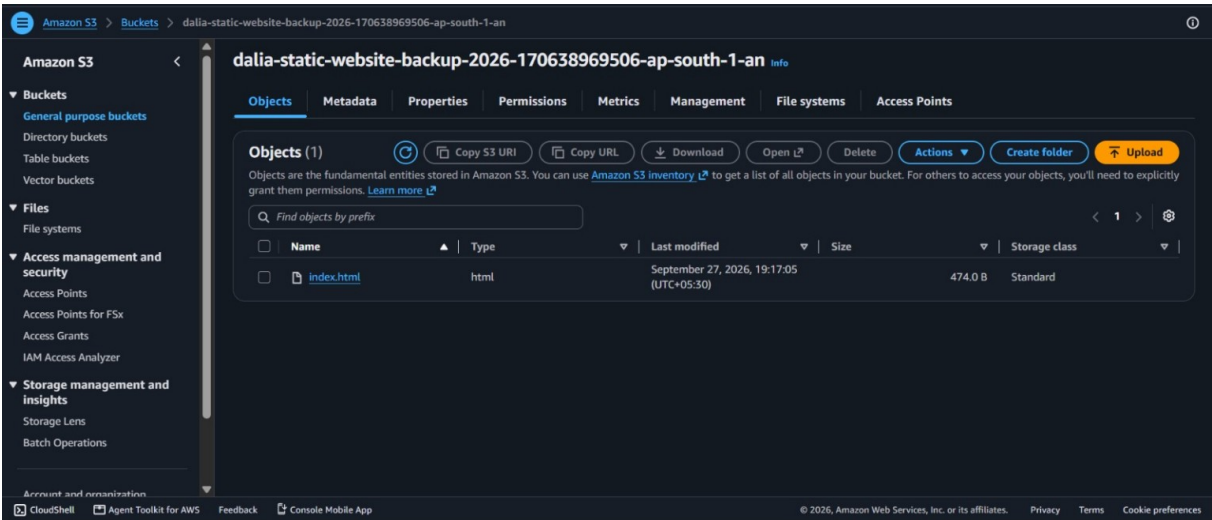


Figure 3: Amazon S3 bucket with index.html

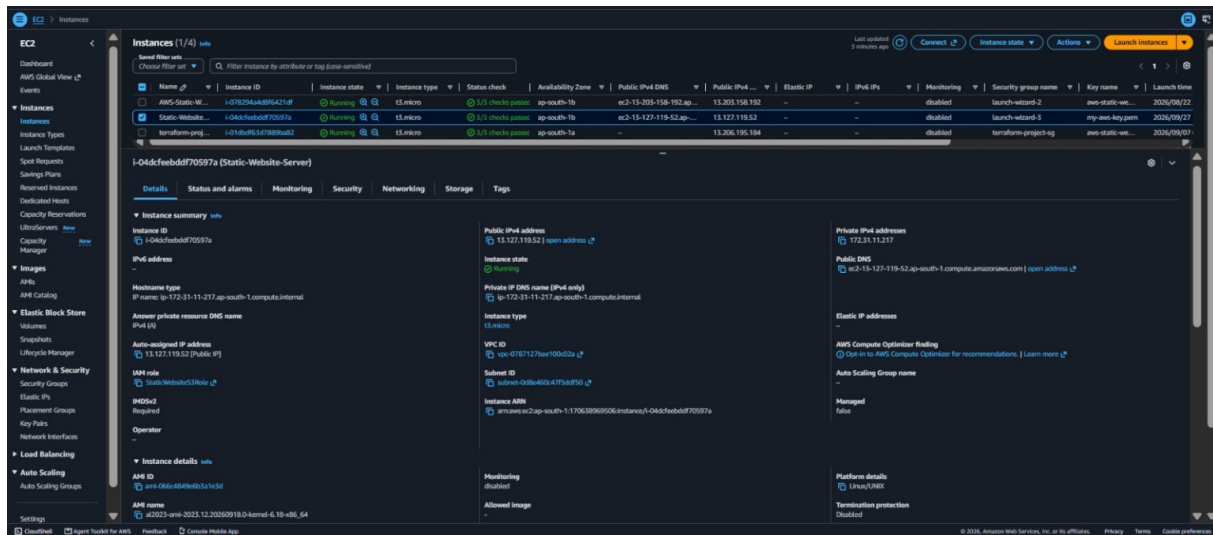


Figure 4: EC2 instance configuration

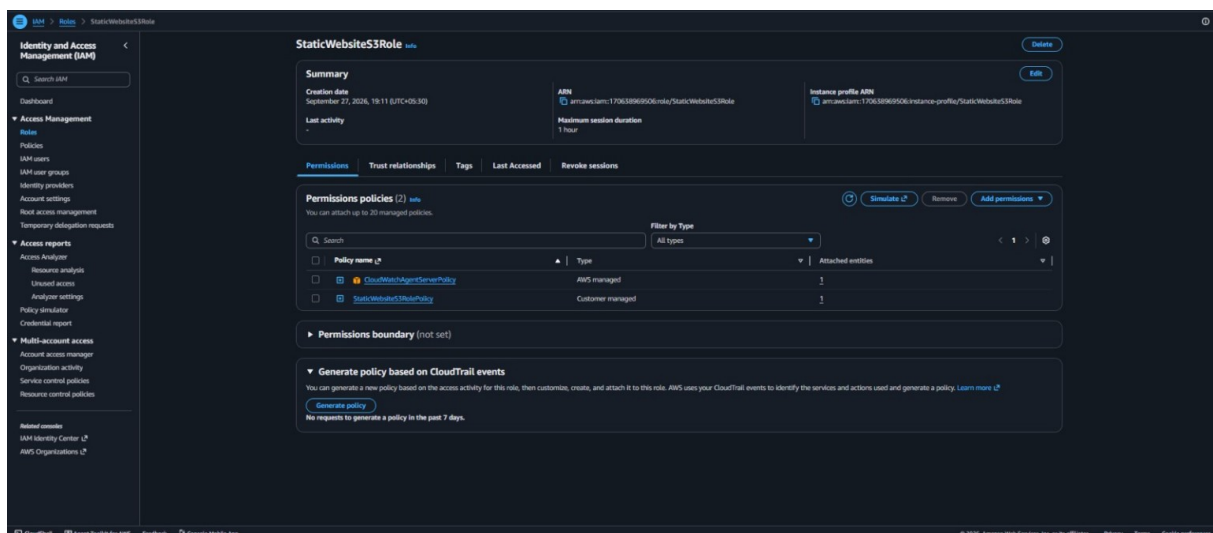


Figure 5: IAM role StaticWebsiteS3Role

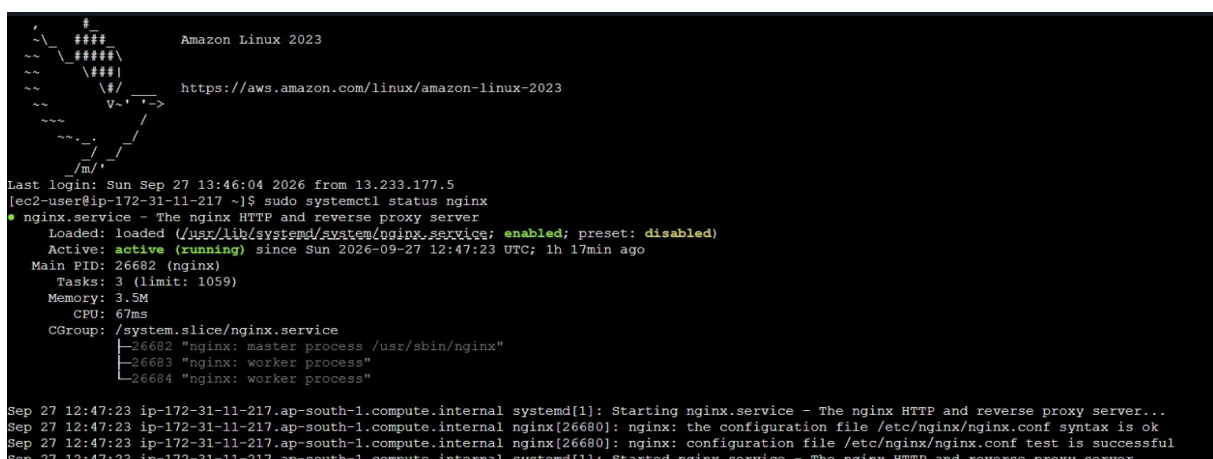


Figure 6: Nginx service status